Thiago Pari

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EDUCATION

Northeastern University May 2027
Master's, Robotics GPA: 3.5
Relevant Coursework: Mobile Robotics, Robotics Sensing and Navigation, Machine Learning, Robot Mechanics

New York University May 2025
Bachelor's, Economics and Mathematics GPA: 3.6
Relevant Coursework: Numerical Analysis, Linear Algebra, ODEs, Complex Analysis

SKILLS AND INTERESTS

Robotics & Simulation: ROS2, MuJoCo, Gazebo, Sensor fusion, Inverse kinematics
AI & LLMs: GPT-4o, Ollama, Function calling, Agentic workflows, ChromaDB, RAG
Prototyping: 3D printing (FDM), Soldering, Fusion 360, SolidWorks, OpenSCAD
Programming: Python, C++, MATLAB, SQL
Tools: Git, Linux (Ubuntu 24.04), Docker
Languages: English, Spanish
Interests: Ironman training (10+ hrs/week: running, cycling, swimming), chess (1600 Elo rapid), guitar

PROJECTS

Northeastern University **Boston, MA**
12-DOF Quadruped Robot with Sim-to-Real Development *October 2025 – Present*

- Designed and fabricated 12-DOF quadruped from CAD (Fusion 360) through 3D printing, integrating Raspberry Pi, servos, PCA9685 PWM driver, and BNO055 IMU via I2C.
- Developing inverse and forward kinematics model in MuJoCo physics simulation, validating gait patterns before hardware deployment through sim-to-real transfer approach.
- Conducted systematic hardware testing including servo calibration, sensor validation, and power system verification.
- Designed multi-voltage power distribution (5V logic, 6V servos) calculating current requirements with safety margin to size power supply appropriately.

Northeastern University **Boston, MA**
AI-Powered Animatronic *October 2025 – Present*

- Built AI assistant integrating hybrid local/cloud LLM architecture (Qwen 2.5 7B via Ollama + GPT-4o-mini), demonstrating 90% cost reduction through intelligent query routing.
- Implemented agentic tool framework with function calling for autonomous web search, browser automation, and code execution.
- Designed custom dual-axis eye mechanism using four-bar linkages in Fusion 360, 3D printing and iterating on tolerances for servo mounts and linkage pins.
- Integrated XIAO ESP32-S3 Sense (built-in camera/mic), PCA9685 servo driver, and I2S audio, demonstrating embedded prototyping.

Northeastern University **Boston, MA**
Multi-Sensor Fusion for Mobile Robot Localization *November 2025 – January 2026*

- Developed ROS2-based Extended Kalman Filter fusing wheel odometry (50Hz), IMU (100Hz), GPS (1Hz), and LiDAR (5-10Hz), achieving 28.8% position error reduction over dead reckoning baseline.
- Built data collection and analysis pipeline with quantitative metrics logging for systematic comparison of filter configurations.
- Created real-time visualization tools in RViz for sensor monitoring and trajectory comparison during testing.

PROFESSIONAL EXPERIENCE

Citi **New York, NY**
Summer Financial Analyst June 2024 – August 2024

- Automated daily reporting workflows by developing Excel VBA macros, reducing processing time by 50% and eliminating manual data collection errors
- Identified non-compliant EUC documentation, implementing quality assurance reviews that reduced Core-IT non-compliance by 34%
- Collaborated cross-functionally with risk management and legal teams to ensure data integrity and regulatory compliance

New York University **New York, NY**
Central Office Intern January 2023 – May 2024

- Managed database operations for 100,000+ entries across Service Link and StarRez systems, ensuring data accuracy and reliability
- Developed efficient workflows for processing student applications, improving response time and service quality for 10,000+ students